FUTURE OF PRIVACY FORUM

bvA webinar - Cookieless – impact op privacy

On post-cookie advertising

June 8, 2021
15:30 – 17:00 CET

Dr. Rob van Eijk
Managing Director for Europe

Future of Privacy Forum

The Supporters

175+	40+	20+	5
Companies	Leading Academics	Advocates and Civil Society	Foundations

The Mission

Privacy leadership and scholarship, advancing principled data practices in support of emerging technologies.
Developing privacy protections, ethical norms, and workable business practices.

The Workstreams

Youth & Education	Ad Tech	Ethics	Global Privacy
Artificial Intelligence	Biometrics	De-identification	Policymaker Education
Mobility Technologies	Smart Communities	Health	AR/VR

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Upcoming Sessions

Understanding Digital Data Flows: De-Identification

July 22, 2021

Understanding Digital Data Flows: Fundamentals of Online Advertising

July 28, 2021

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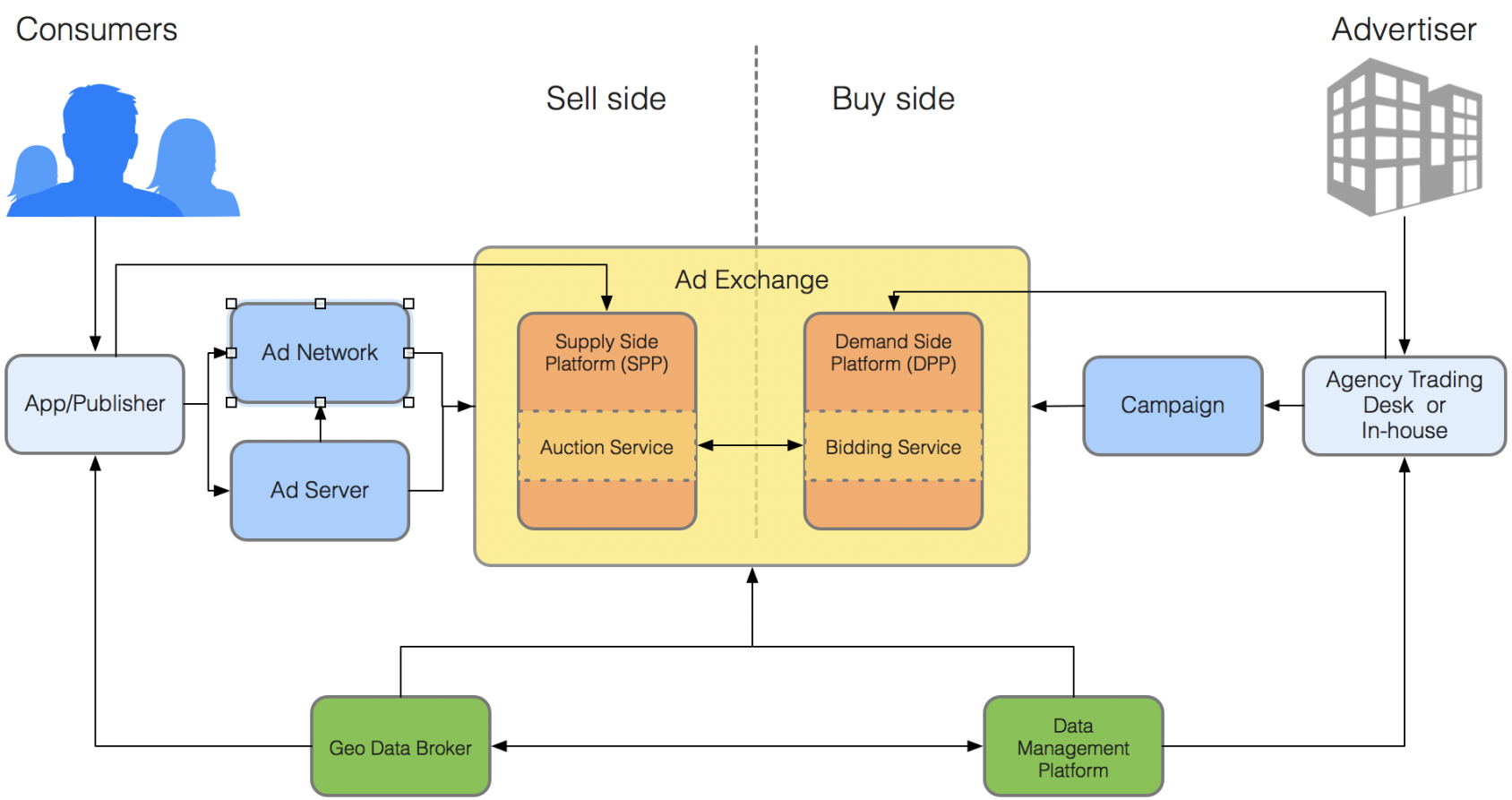
Ad Tech Value Chain



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Contemporary Ad Models

Targeting Types	Example:
Contextual tailored to a website or category of websites	An **airline** buys ads on **travel-related websites**.
Behavioral third party intermediaries create “audience segments” based on users’ activities across many sites and platforms	An advertiser buys categories (segments) of viewers, such as “intending to buy a car,” “back to school shopping,” or “wedding planning.”
Re-targeting first party targets ads to reach its own previous visitors or customers	Retail site places a “tag” on a product page, and re-targets users who viewed that page with ads elsewhere on the internet.

Pricing Models:

- **Cost Per Click (CPC) –** how many users clicked?
- **Cost per (1k) Impressions (CPM) –** how many users saw the ad (impressions)?
- **Cost-Per-Action (CPA)** – how many users took an action…

  … visited a site? … downloaded an app? … made a purchase? … entered a store?

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Cookie Differences By Browser

	Chrome	Firefox	Safari
Cookies in a third-party context	No restrictions*	Access restricted for known trackers*	All access restricted, except with Storage Access API
	* Google Chrome will no longer support 3rd party cookies **sometime in 2022**.	* Firefox added (1) SmartBlock to preserve functionality and (2) cookie silos to prevent sharing cookies between websites	
Protection mechanism	N/A	Enhanced Tracking Protection (Total Cookie Protection)	Intelligent Tracking Prevention (ITP)
Default protection mode	N/A	By Default	By Default
Classification of "known trackers"	N/A	Disconnect.me	Algorithmic

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Source: https://www.cookiestatus.com/

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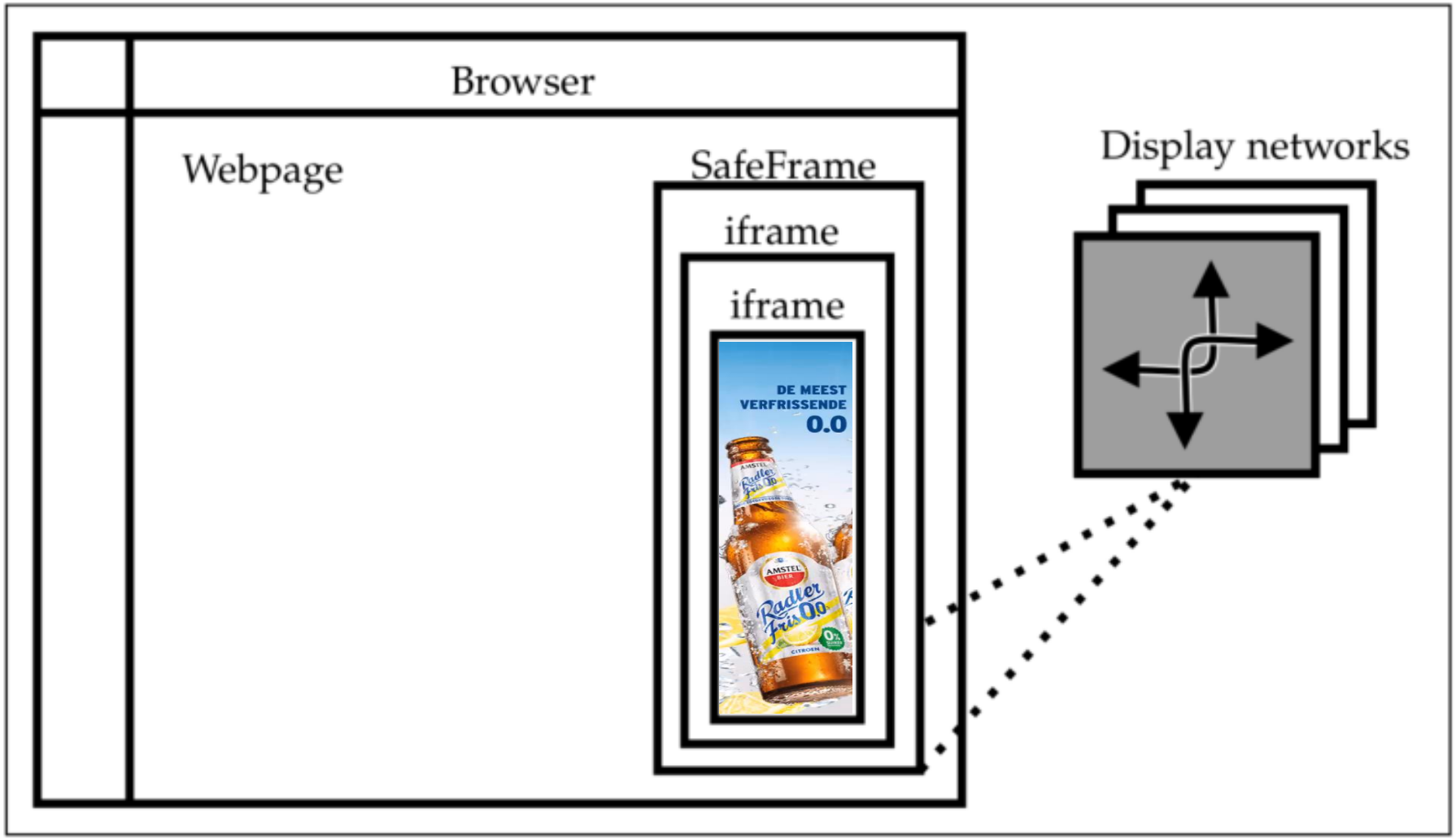
Behind the Scenes of an Online Ad



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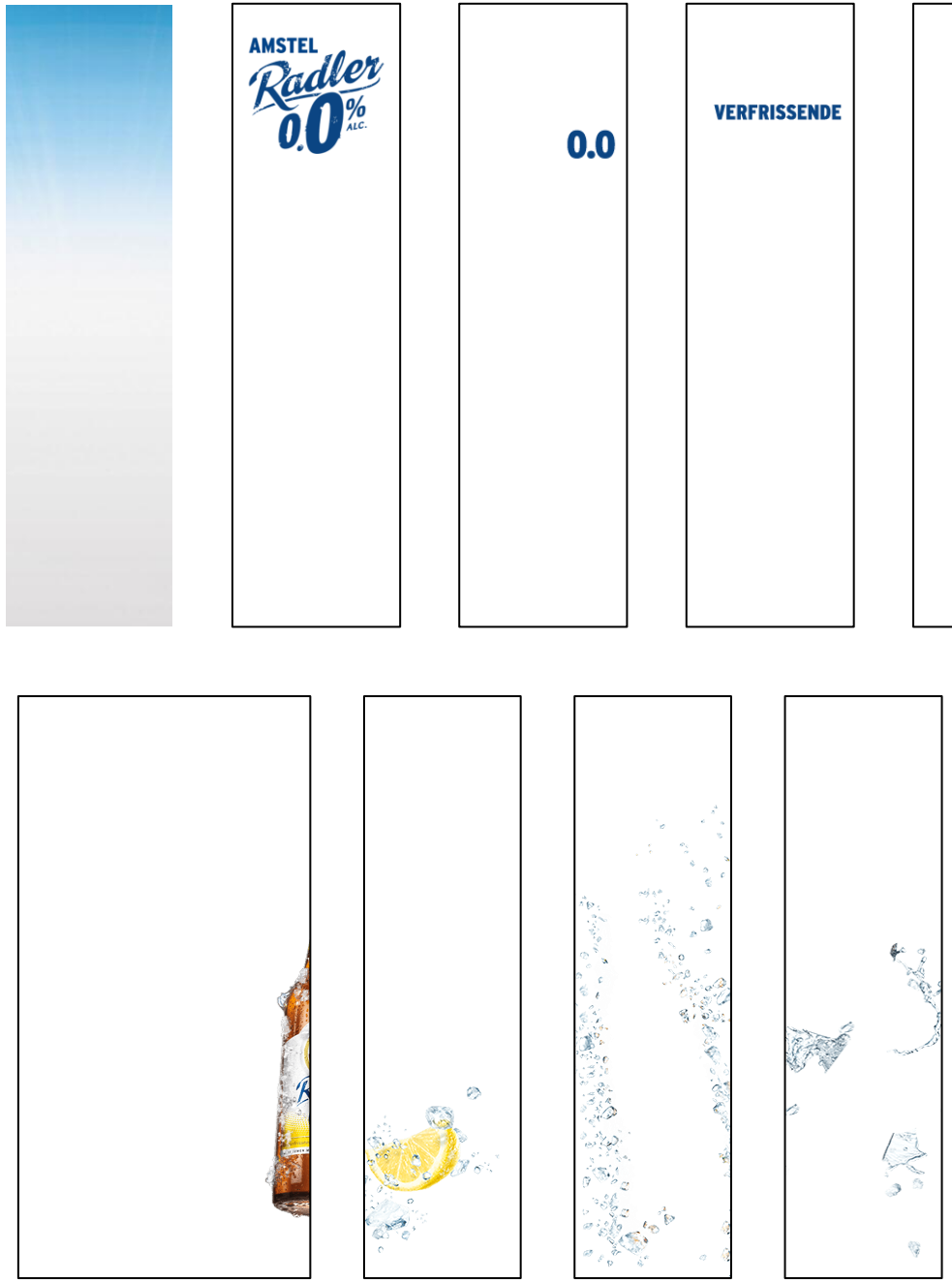

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2mdn.net								
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Google Doubleclick Frame:
https://s0.2mdn.net/9026094/1559035272353/amnet_160x600/index.html

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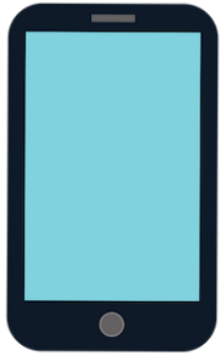
Probabilistic

- Involves making inferences, using a variety of “signals” or data points, that multiple devices ***probably*** belong to the same person.
- **Example**: IP Address or Local Router SSID (multiple devices accessing the same home Wi-Fi network)

A user visits a website on a **computer** and a mobile **device**



Cookie ID

Mobile Ad ID

Home WiFi

Data Algorithm Partner

Mapped IP Address

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Probabilistic Alternative to cookies: Server-Side Recognition (Fingerprinting)

Creation of a statistical identifier or “fingerprint” for a browser

Purposes: security, authentication, and/or recognizing users over time for advertising and reporting if cookies are not available

- Browser is queried for its agent string, screen color depth, language, installed plugins, time zone, etc.
- Few browsers will share fingerprints (in the example to the right, 1 in 280,000).
- May lose effectiveness over time.
- Challenging to effectuate opt outs

userAgent	// "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/60.0.3112.113 Safari/537.36"
language	// “en-US”
plugins	"Plugin 0: Chrome PDF Viewer; Plugin 1: Chrome PDF Viewer; Portable Document Format; internal-pdf-viewer. Plugin 2: Native Client; ; internal-nacl-plugin. Plugin 3: Shockwave Flash; Shockwave Flash 27.0 r0; pepflashplayer.dll. Plugin 4: Widevine Content Decryption Module; Enables Widevine licenses for playback of HTML audiovideo content. version: 1.4.8.1000; widevinecdmadapter.dll."
platform	// “Win32”
do not track	// “yes”
timezone	// “240”
screen resolution	// “1920x1080x24”

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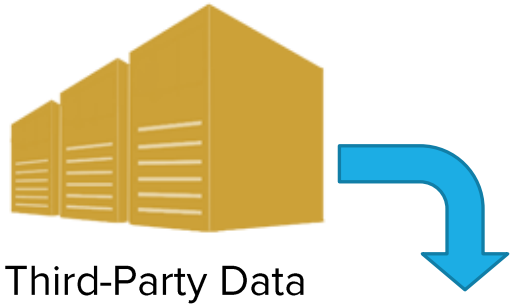
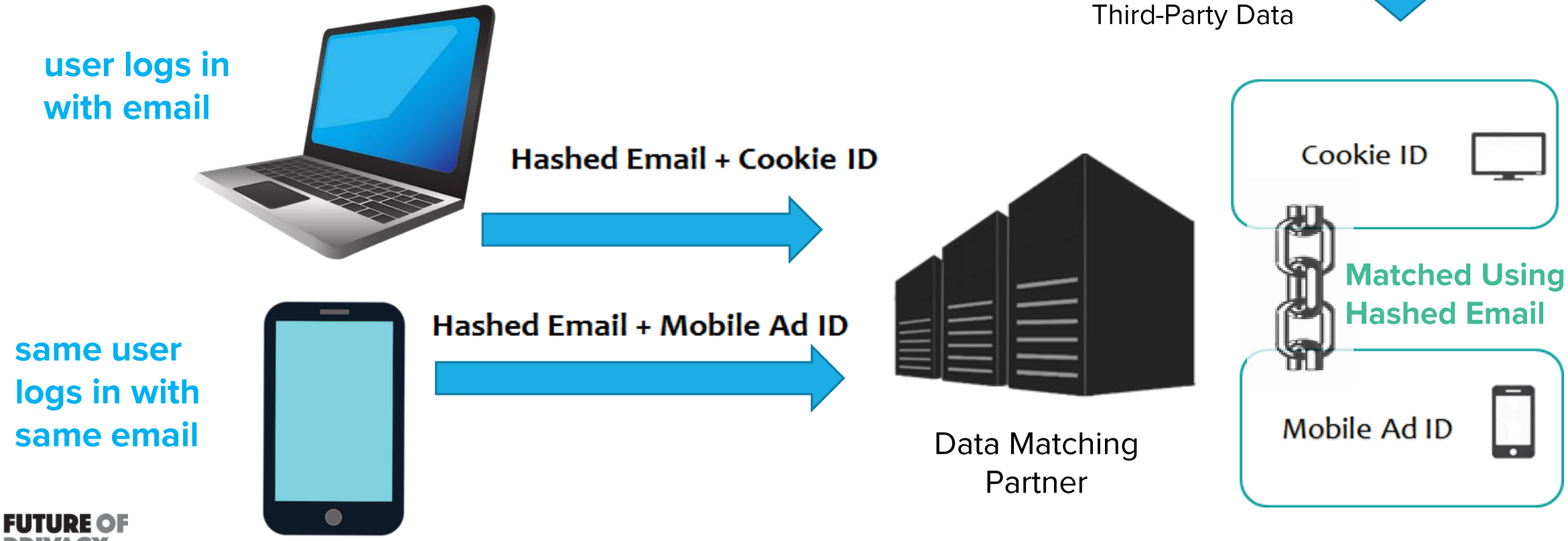
Deterministic

- Involves tracking a unique, stable identifier, such as an email address, when it is used across multiple **devices** to log into apps and websites.





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Multiple discussions around multiple post-cookie solutions

- Four examples of groups that discuss post-cookie solutions:
  - PRAM (Partnership for Responsible & Addressable Media)
  - Prebid.org
  - IAB (Interactive Advertising Bureau)
  - W3C (World Wide Web Consortium)
- PRAM Business Use cases

https://docs.google.com/document/d/17geRxY4Wko_yzLkezlbGMIeD7s-DcaJOQfsCMMw8lxE/edit

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Multiple discussions around multiple post-cookie solutions

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  - Prebid.o
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  - W3C (World Wide Web Consortium
- PRAM Business Use cases
- W3C proposals aka the Privacy Sandbox

https://docs.google.com/document/d/17geRxY4Wko_yzLkezlbGMIeD7s-DcaJOQfsCMMw8lxE/edit

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bslassey / ip-blindness

Code | Issues 2 | Pull requests | Actions | Projects | Wiki | Security | Insights

master | 1 branch | 0 tags | Go to file | Code

bslassey Adding Gnatcatcher proposal a3540d6 on Jan 13 9 commits

LICENSE.md	Create LICENSE.md	2 years ago
README.md	Adding Gnatcatcher proposal	5 months ago
near_path_nat.md	fix link and bold for real	5 months ago
willful_ip_blindness.md	Adding Gnatcatcher proposal	5 months ago

README.md

Gnatcatcher

Global Network Address Translation Combined with Audited and Trusted CDN or HTTP-Proxy Eliminating Reidentification

Background

We have previously proposed two separate solutions for IP Blindness: Near-Path NAT and Willful IP Blindness

About

No description, website, or topics provided.

Readme

Releases

No releases published

Packages

No packages published

Contributors 2

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JensenPaul Paul Jensen

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bslassey / ip-blindness

Code Issues 2

Federated Learning of Cohorts (FLoC)

This is an explainer for a new way that browsers could enable interest-based advertising on the web, in which the companies who today observe the browsing behavior of individuals instead observe the behavior of a cohort of similar people.

Overview

The choice of what ads to show on a web page may typically be based on three broad categories of information: (1) First-party and contextual information (e.g., "put this ad on web pages about motorcycles"); (2) general information about the interests of the person who is going to see the ad (e.g., "show this ad to Classical Music Lovers"); and (3) specific previous actions the person has taken (e.g., "offer a discount on some shoes that you left in a shopping cart"). **This document addresses category (2), ads targeting based on someone's general interests.** For personalized advertising in category (3), please check out the TURTLEDOVE proposal.

In today's web, people's interests are typically inferred based on observing what sites or pages they visit, which relies on tracking techniques like third-party cookies or less-transparent mechanisms like device fingerprinting. It would be better for privacy if interest-based advertising could be accomplished without needing to collect a particular individual's browsing history.

Background

We have previously proposed two separate solutions for IP Blindness: Near-Path NAT and Willful IP Blindness

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User-Agent Client Hints

Draft Community Group Report, 2 June 2021

This version:

https://wicg.github.io/ua-client-hints/

Editors:

Mike Taylor (Google LLC)
Yoav Weiss (Google LLC)

Former Editor:

Mike West (Google LLC)

Participate:

File an issue (open issues)

Copyright © 2021 the Contributors to the User-Agent Client Hints Specification, published by the Web Platform Incubator Community Group under the W3C Community Contributor License Agreement (CLA). A human-readable summary is available.

Abstract

This document defines a set of Client Hints that aim to provide developers with the ability to perform agent-based content negotiation when necessary, while avoiding the historical baggage and passive fingerprinting surface exposed by the venerable `User-Agent` header.

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Code | Issues 2

User-Agent Client Hints

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WICG / conversion-measurement-api

Notifications

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main | 26 branches | 0 tags | Go to file | Code

csharrison Add 2021-05-17 minutes — 6c0bbc8 3 hours ago — 182 commits

meetings	Add 2021-05-17 minutes	3 hours ago
static	Add updated proposal for aggregate conversions	14 months ago

About

Conversion Measurement API

wicg.github.io/conversion-measureme...

Readme

View license

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Abstract

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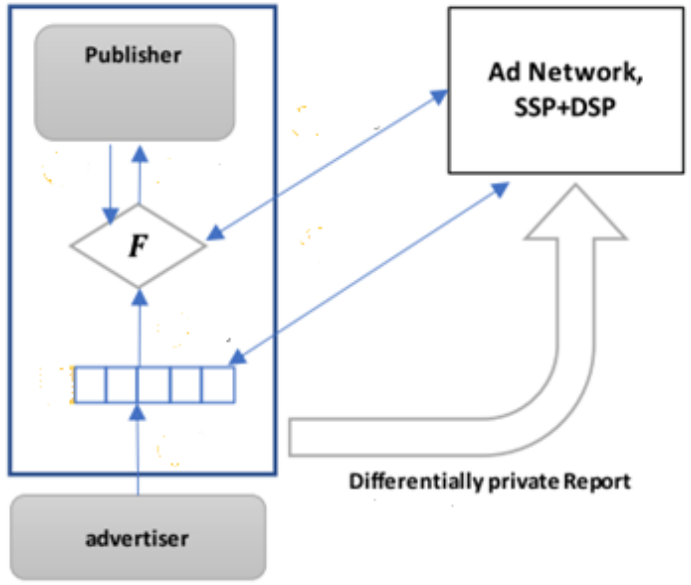
Code Issues 2

User-Agent Client Hints

Draft Community Group Report 8 June 2021

TURTLEDOVE



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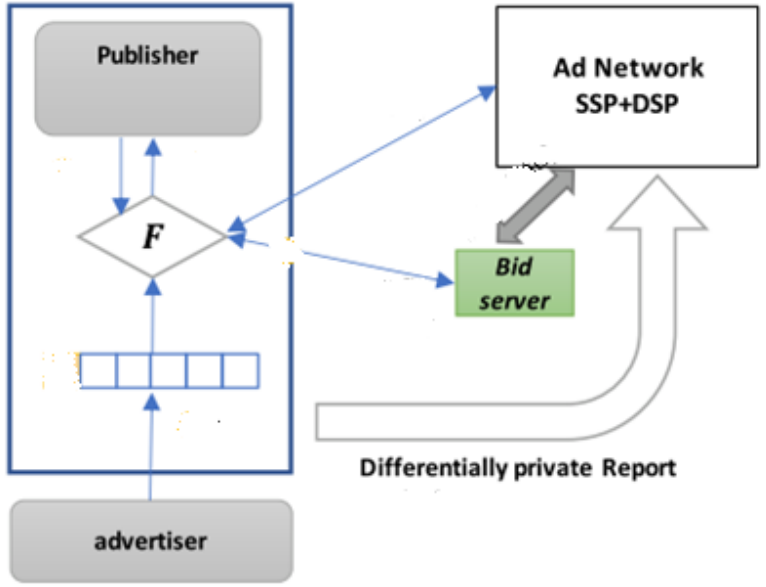

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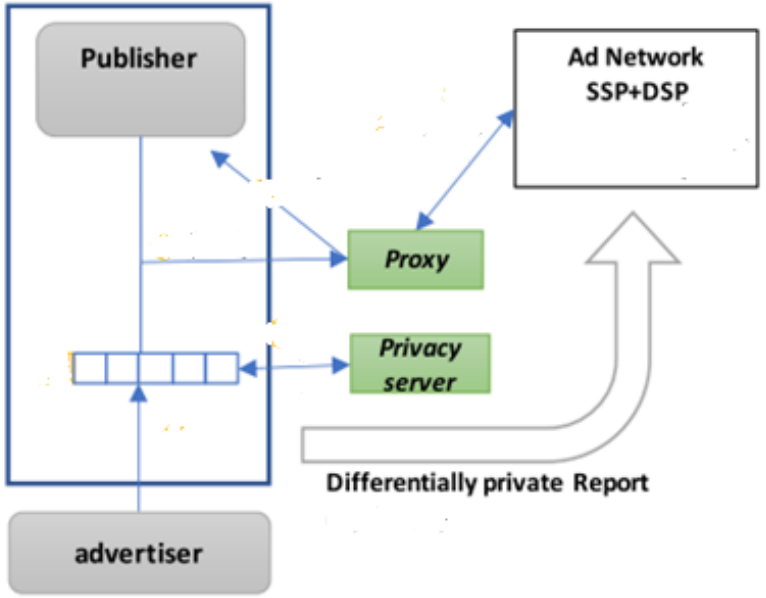

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What to expect in terms of adtech and value chain?

- A patchwork of federated adtech solutions
- 1st parties will be more influential
- Intermediairies will be less relevant
- More transparency for and control by data subjects
- Clean rooms
  - To anonymize the context provided by the publisher
  - To anonymize the user's geographic location :
- Expect a patchwork of cryptographic solutions
  - Differential privacy
  - Homomorphic encryption
  - Secure multi-party computation (privacy-preserving computation)
  - **Privacy budget** constraints for clean rooms

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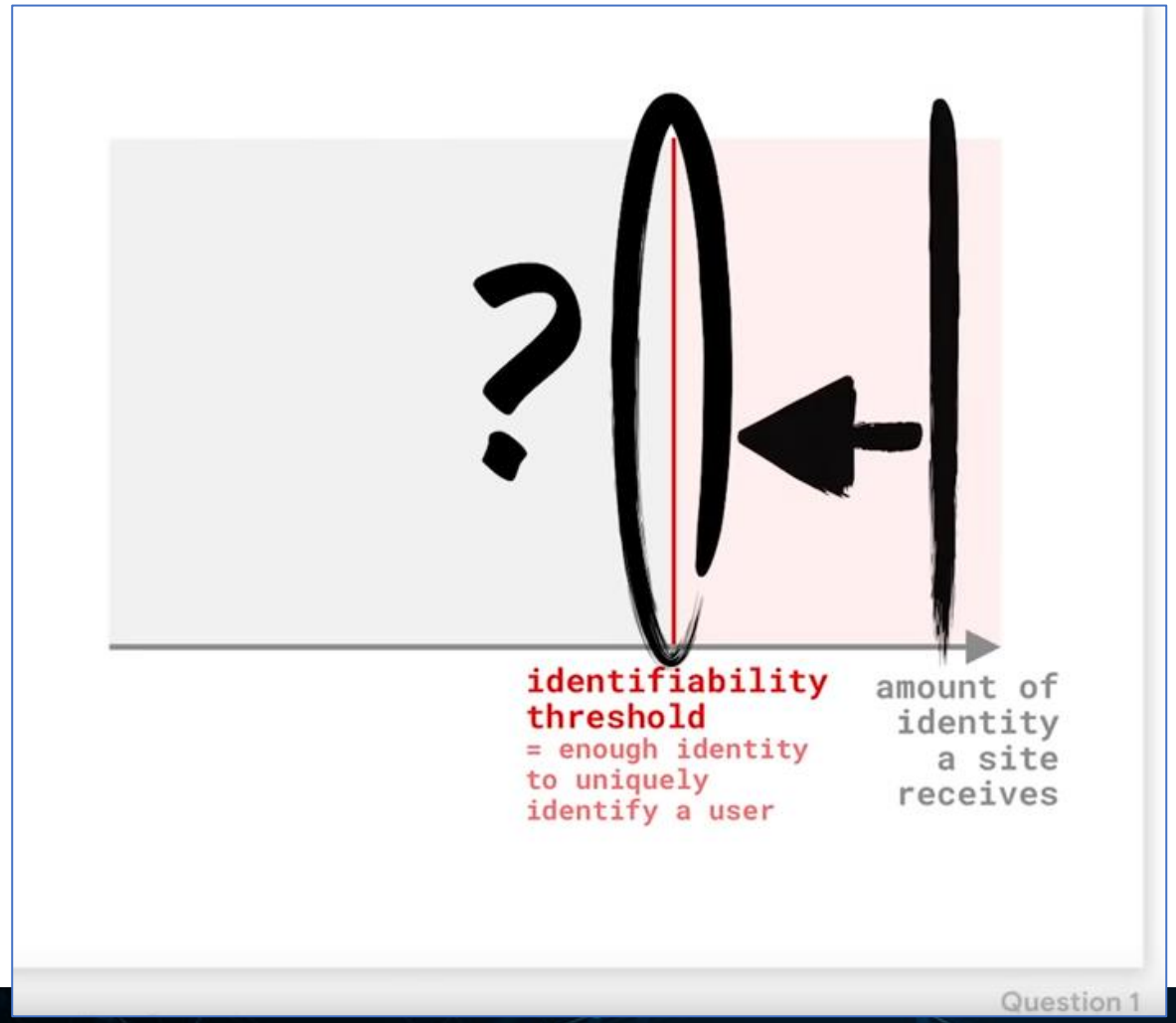
How to make sense of the Privacy Budget?



https://www.youtube.com/watch?v=0STgfjSA6T8

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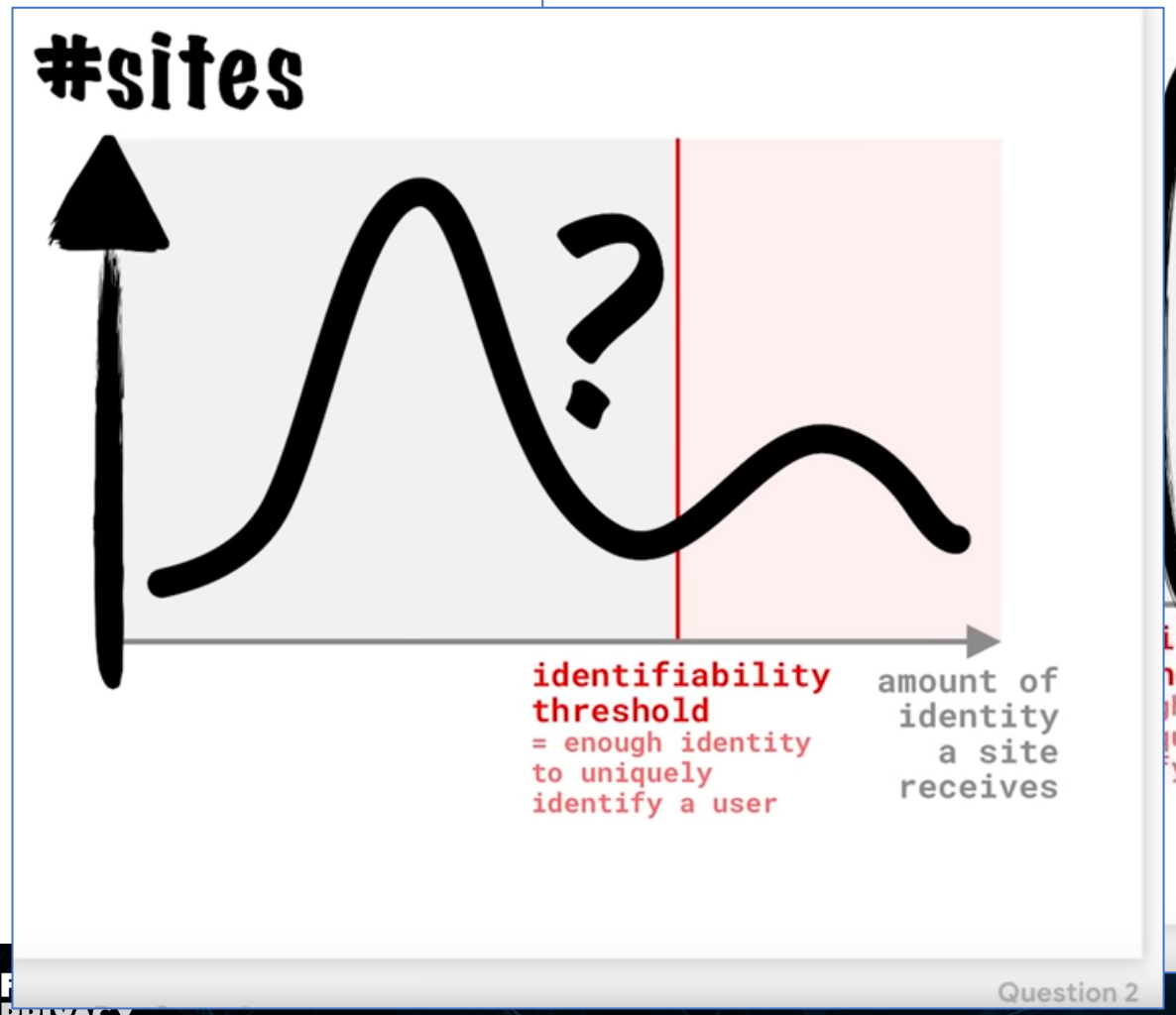
How to make sense of the Privacy Budget?



https://www.youtube.com/watch?v=0STgfjSA6T8

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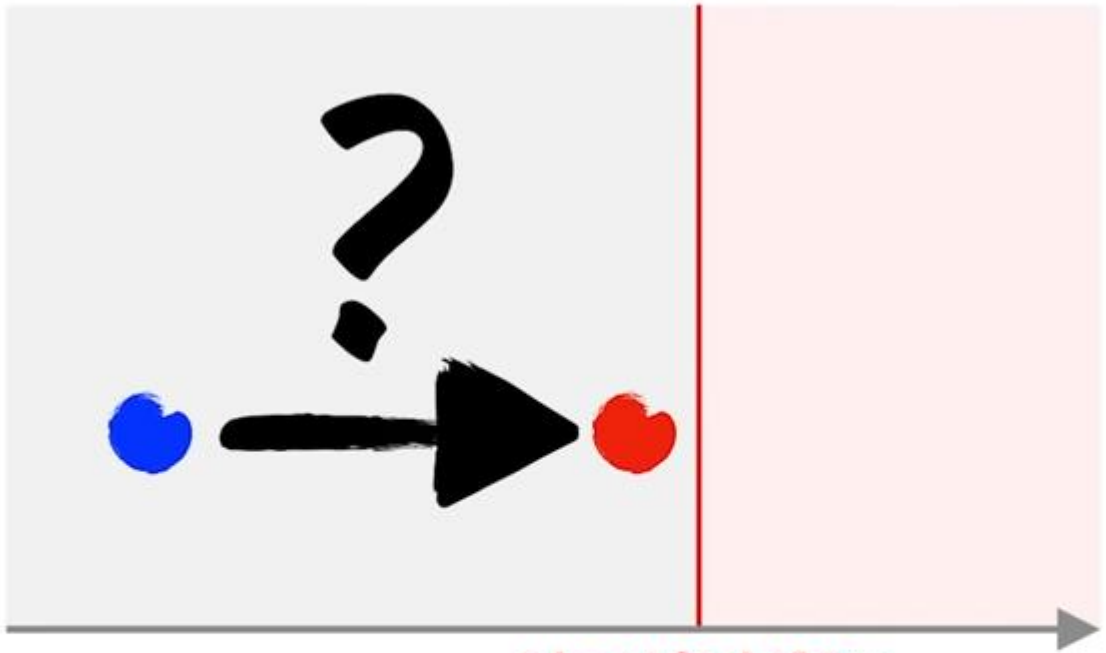
300+ APIs

- Local fonts
- Canvas output
- User-Agent
- Audio, extensions...

font1 font2 *font3*

```
Mozilla/5.0 (X11; Linux
x86_64) AppleWebKit/537.36
(KHTML, like Gecko) Chrome...
```



Question 3

Question 1

Question 2

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In summary: Privacy Sandbox

How identifying each API is

How each API correlates with others 🤯

= How identifying subsets of APIs are

Which subsets of APIs sites are using

= How much identity is leaked to each site

Chrome Dev Summit 2020

amount of identity a site receives

Question 3

Question 1

https://www.youtube.com/watch?v=0STgfjSA6T8

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What data will be available for post cookie advertising?

- Contextual data
- Household level data (linkable to the IP address as a proxy for location)
- First-party and second-party data
- Identity-based data (although input data is impacted and new linkages will be needed when stored in a graph)
- Federated data & cohorts

https://www.adexchanger.com/data-exchanges/6-types-of-post-cookie-data-that-will-still-be-available-after-2022/

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The impact of the consent banner on ad data flows

- It should be as easy to revoke consent as it is to give consent!

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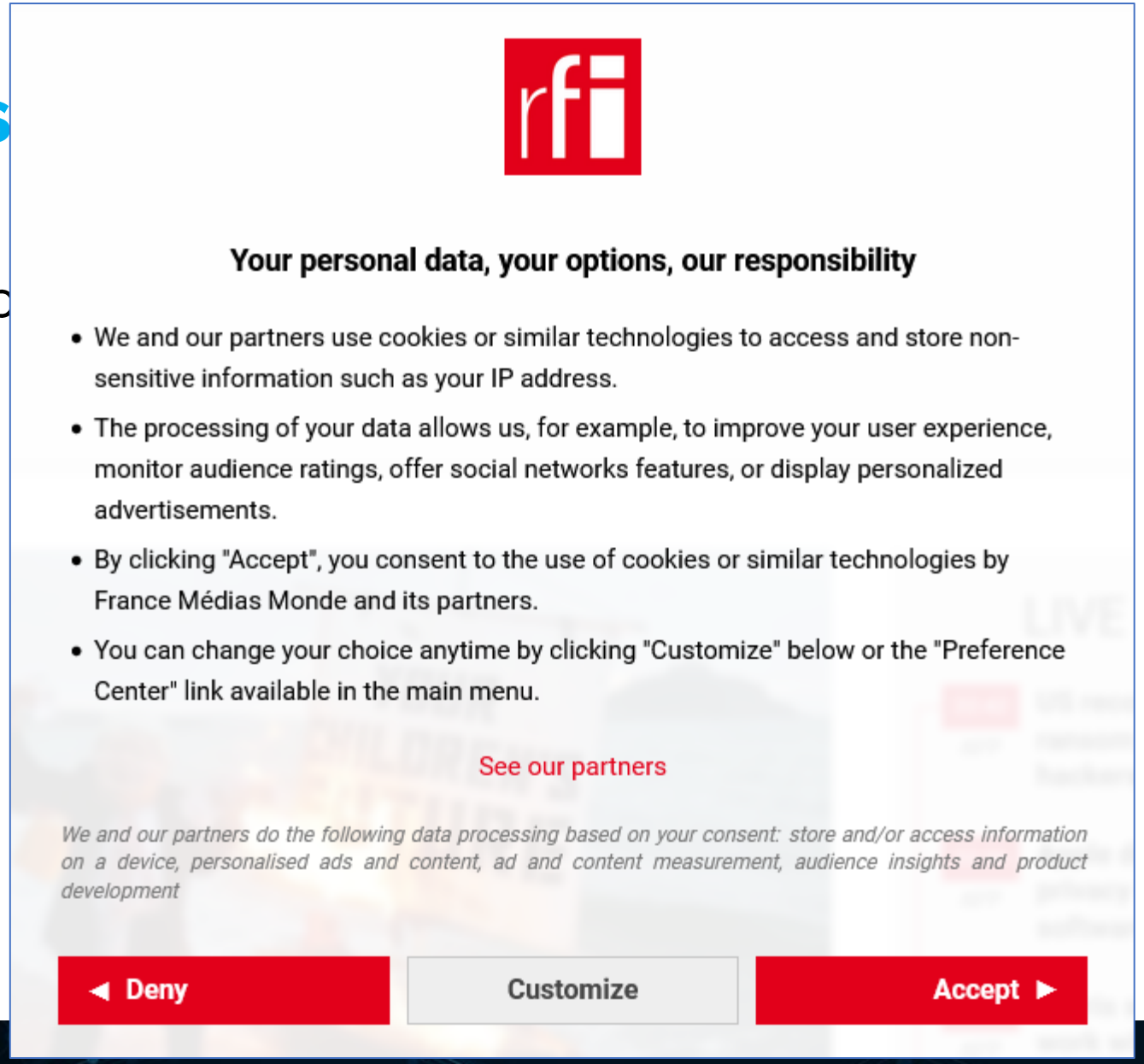
The impact of the cons

- It should be as easy to revoke co



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Click below to consent to the above or make granular choices, including exercising your right to object to companies processing personal data based on legitimate interest instead of consent. You can update your choices at any time by clicking on the Privacy icon in the bottom of the screen.

We use your data for:

Precise geolocation data, and identification through device scanning

Personalised ads and content, ad and content measurement, audience insights and product development

Functional

Our Partners

Reject All | Manage Settings

Accept All

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Your personal data, your options, our responsibility

- We and our partners use cookies or similar technologies to access and store non-sensitive information such as your IP address.
- The processing of your data allows us, for example, to improve your user experience, monitor audience ratings, offer social networks features, or display personalized advertisements.
- By clicking "Accept", you consent to the use of cookies or similar technologies by France Médias Monde and its partners.
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See our partners

We and our partners do the following data processing based on your consent: store and/or access information on a device, personalised ads and content, ad and content measurement, audience insights and product development

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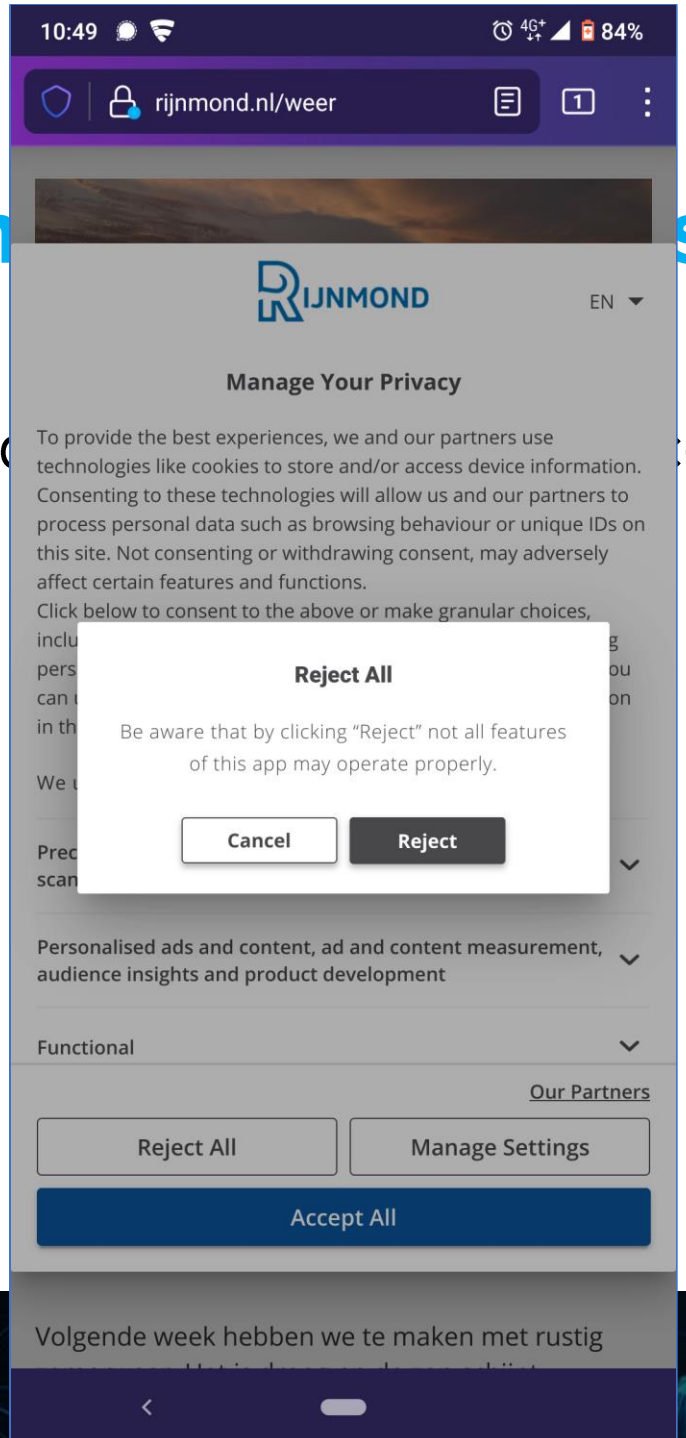

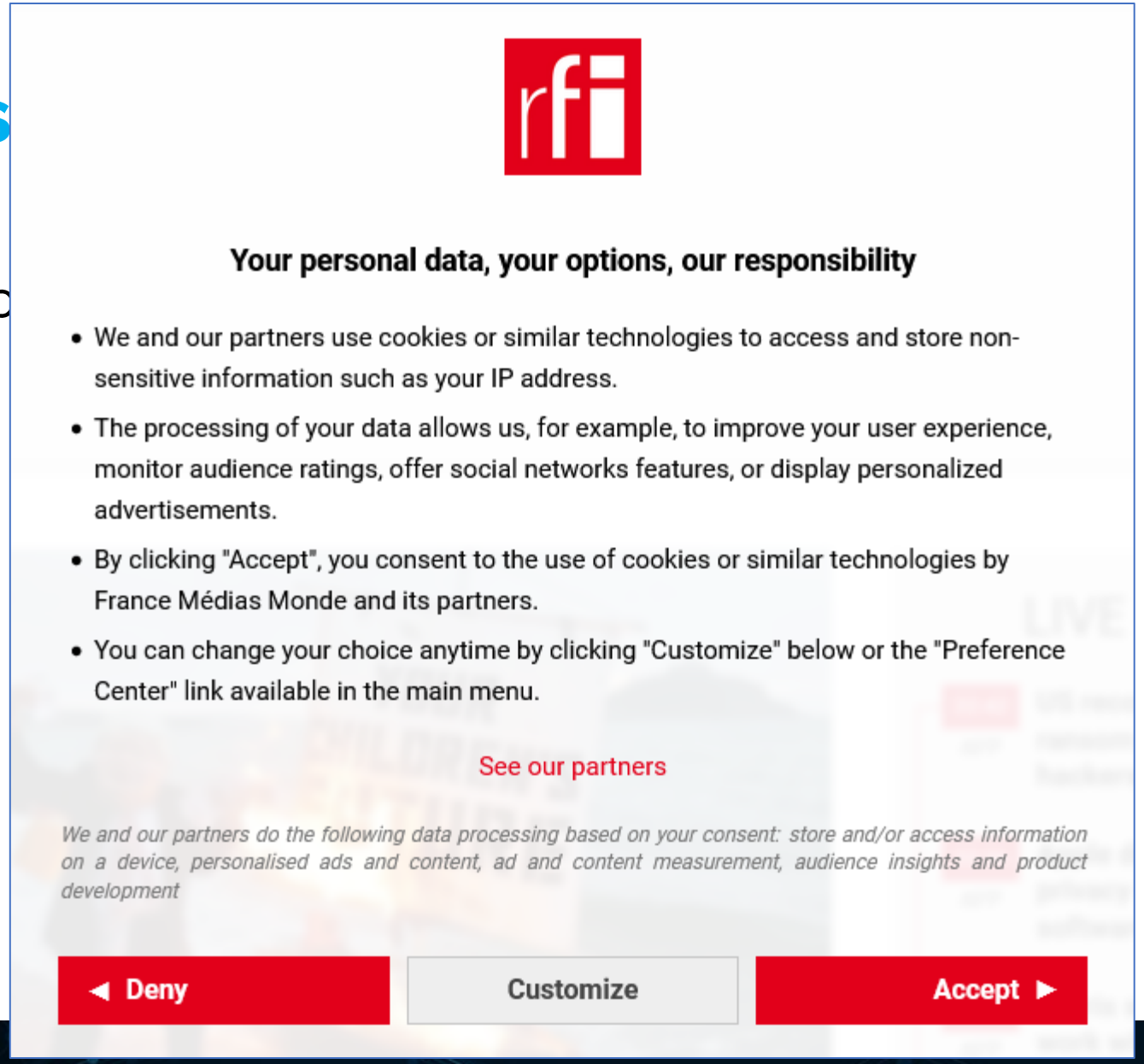

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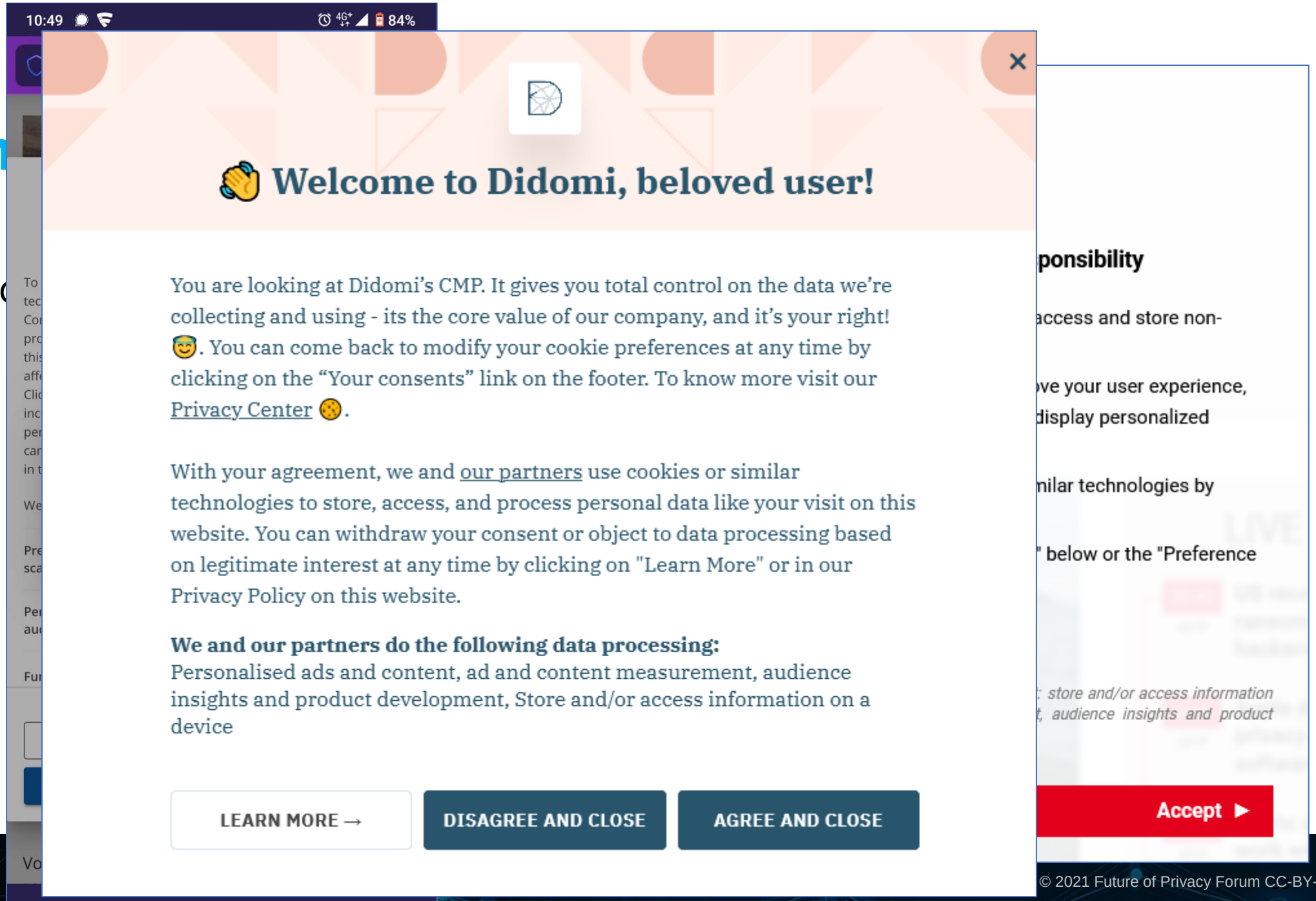

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So, you want to find out more?

Data privacy seems important to you - it's really important for us too! In fact, we're specialized in it 🤓. We keep cookie data for 13 months. You can accept or reject all cookies at once, by using the buttons at the bottom of this window.

We use this data to understand you as a user, improve your experience on didomi.io and to show you personalised ads. If you want to know more, you can explore our product below or visit our **Privacy Center**.

- Storing and/or accessing information on a terminal

DISAGREE AGREE

Cookies, identifiers of your terminal or other information may be stored or consulted on your terminal for the purposes presented to you.

- **Store and/or access information on a device**

Cookies, device identifiers, or other information can be stored or accessed on your device for the purposes presented to you.

Consent DISAGREE AGREE

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: store and/or access information
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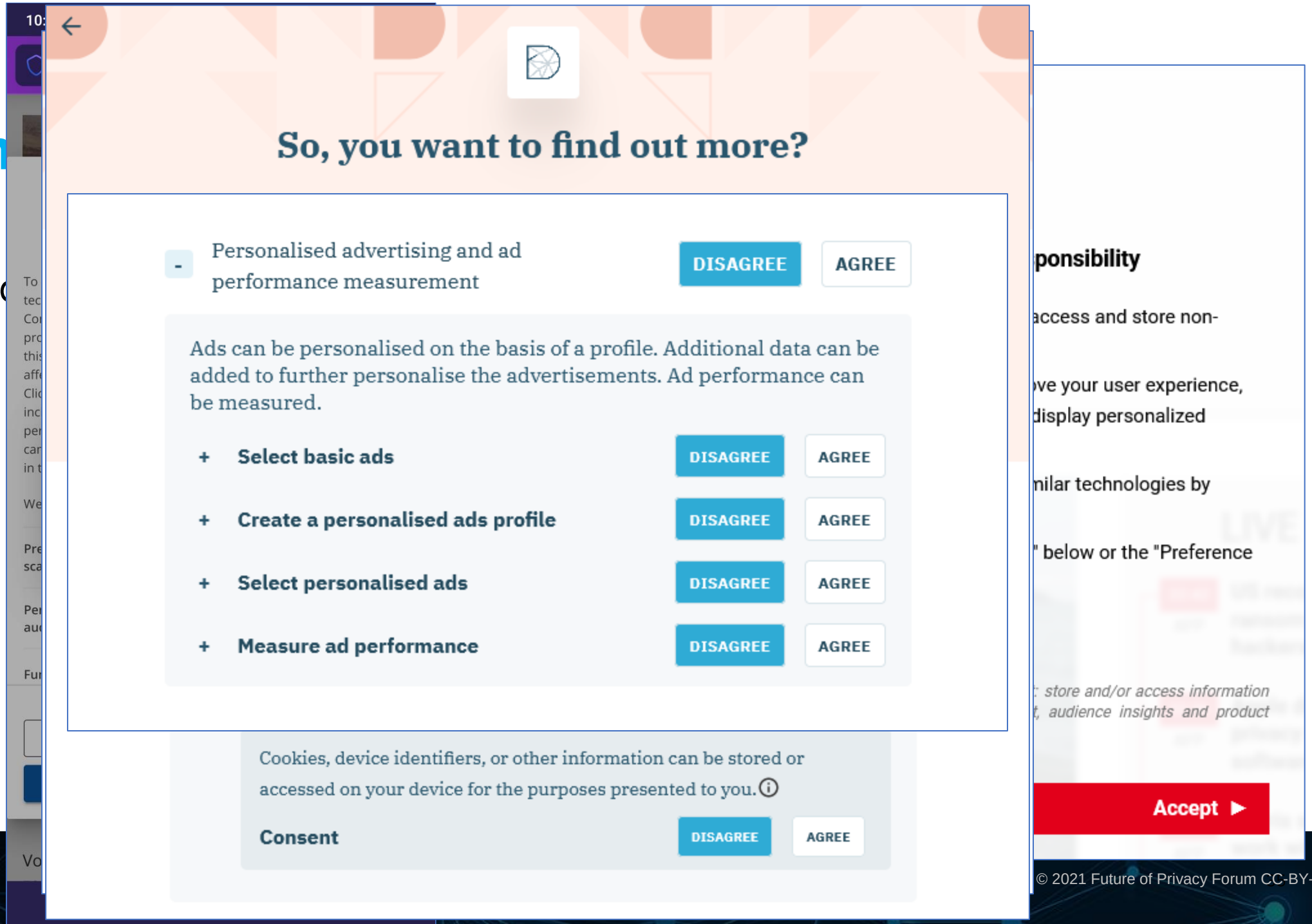

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Personalised content, content performance measurement, audience data, and product development

DISAGREE AGREE

The content can be personalised on the basis of a profile. Additional data can be added to further personalise the content. Content performance can be measured. Information can be generated on which audiences have seen the ads and content. Data can be used to create or improve user experience, systems and software.

+ Create a personalised content profile	DISAGREE	AGREE
+ Select personalised content	DISAGREE	AGREE
+ Measure content performance	DISAGREE	AGREE
+ Apply market research to generate audience insights	DISAGREE	AGREE
+ Develop and improve products	DISAGREE	AGREE

By giving consent to the purposes above, you also allow this website and its partners to operate the following data processing: Ensure security, prevent fraud, and debug, Link different devices, Match and combine offline data sources, and Technically deliver ads or content

By none of our partners VIEW OUR PARTNERS

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Upcoming Sessions

Understanding Digital Data Flows: De-Identification

July 22, 2021

Understanding Digital Data Flows: Fundamentals of Online Advertising

July 28, 2021

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Questions?

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